Untitled

```r
library(tidyverse)

n <- 1000

X <- runif(n,0,1)

Y <- 10*X + rnorm(n,0,1)

b0 <- 0

b1 <- 1/mean(X^2)

((b1*X + b0)*Y) |> mean()
```

```
[1] 9.999052
```

Estimand is

$$\theta = \int \frac{\partial E[Y \mid X]}{\partial X} f(X) dX$$

$$= 1$$

1 Riesz representation

$$\min \sum \left(\alpha(X)^2 - 2 \frac{\partial \alpha(X)}{\partial X} \right)$$

With linear weight $\alpha(X) = \beta_0 + \beta_1 X$,

$$\min \sum \left([\beta_0 + \beta_1 X]^2 - 2\beta_1 \right)$$

First order condition is

$$0 = \sum [\beta_0 + \beta_1 X]$$

$$0 = \beta_0 + \beta_1 \times E[X]$$

and

$$0 = \sum ([\beta_0 + \beta_1 X] X - 1)$$

$$1 = \beta_0 E[X] + \beta_1 E[X^2]$$

Then,

$$1 = -\beta_1 E[X]^2 + \beta_1 \times E[X^2]$$

$$\frac{1}{E[X^2] - E[X]^2} = \beta_1$$

$$\beta_0 = -\frac{E[X]}{E[X^2] - E[X]^2}$$